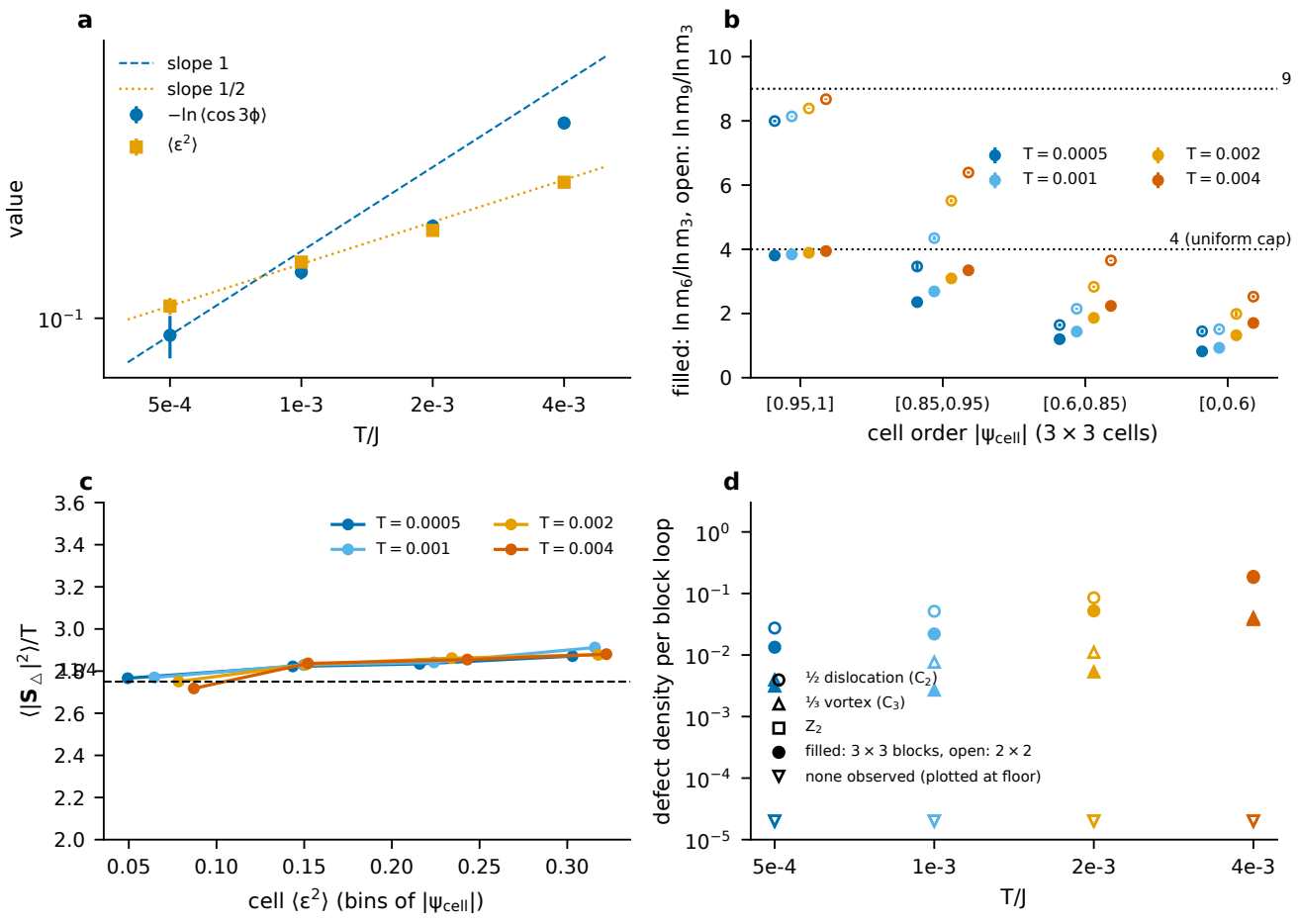
a
slope 1
slope 1/2
−ln ⟨cos 3ϕ⟩
⟨ε²⟩
value
10⁻¹
5e-4
1e-3
2e-3
4e-3
T/J
b
filled: ln m₆/ln m₃, open: ln m₉/ln m₃
10
8
6
4
2
0
9
4 (uniform cap)
T = 0.0005
T = 0.001
T = 0.002
T = 0.004
[0.95,1]
[0.85,0.95)
[0.6,0.85)
[0,0.6)
cell order |ψcell| (3 × 3 cells)
c
⟨|S△|²⟩/T
3.6
3.4
3.2
3.0
2.8
2.6
2.4
2.2
2.0
T = 0.0005
T = 0.001
T = 0.002
T = 0.004
0.05
0.10
0.15
0.20
0.25
0.30
cell ⟨ε²⟩ (bins of |ψcell|)
d
defect density per block loop
10⁰
10⁻¹
10⁻²
10⁻³
10⁻⁴
10⁻⁵
½ dislocation (C₂)
⅓ vortex (C₃)
Z₂
filled: 3 × 3 blocks, open: 2 × 2
none observed (plotted at floor)
5e-4
1e-3
2e-3
4e-3
T/J